



Figure 4: Head-to-head comparison results of BoN on the STORYRMB. “Tie” means both models select the same story.

H ADDITIONAL EXPERIMENTS

H.1 ABLATION STUDY

The results of the ablation experiment are presented in Table 2. We analyze the impact of removing key components from the method of constructing the training dataset.

(-Premise Back-Generation): We generate two stories from the same prompt using two models with different scales, where the story generated by the larger one is treated as the chosen story. These stories are seemed as supplementary training data. We then remove those training data generated by “Premise Back-generation” and add the supplementary data of the same quantity. We then leverage Qwen3-8B as the base models for training and evaluate the results on STORYRMB.

(-Human-Guided Continuation): Similarly, we remove the training data generated by “Human-Guided Continuation” and replace it with supplementary training data of the same quantity. We also leverage Qwen3-8B as the base models for training and evaluate the results on STORYRMB.

(-Prompt-Guided Rewriting): Be the same as the method before, we remove the training data generated by “Prompt-Guided Rewriting” and replace it with supplementary training data of the same quantity. We leverage Qwen3-8B as the base models for training and evaluate the results on STORYRMB.

H.2 HEAD-TO-HEAD RANKING RESULTS ON STORYRMB

We further conduct an evaluation using the head-to-head ranking evaluation method in § 5 on STORYRMB. In this setting, we compare the ranking of selected stories of two reward models, where the winning story is the one ranked higher in the original preference ranking. The results are shown in Figure 4. We can observe that our model outperforms baselines, which is consistent with the findings in Figure 3. Regarding the lag LLM-LLM pairs in Figure 2, one possible reason is that while our model excels at general preference, distinguishing the absolute best candidate among high-quality, similar LLM generations remains a challenge.

H.3 PERFORMANCE ON ANOTHER WRITING BENCHMARK

We evaluate our models on another writing benchmark. Specifically, we use the Creative Writing v3 evaluation set of EQ-Bench (Paech, 2023) with a widely-used evaluation method for story (Chhun et al., 2024). We first conduct a meta-evaluation of this evaluation method on STORYRMB, i.e., using this method to score each story and select the best one. We find it achieves 94% accuracy on LLM-LLM pairs and 54% on human-LLM pairs. Therefore, we believe this evaluation method is effective to evaluate LLM-generated stories.

Table 2: “(-) Premise Back-generation” removes training data constructed by “Premise Back-generation”. “(-) Prompt-Guided Rewriting” removes training data constructed by “Prompt-Guided Rewriting”. “(-) Human-Guided Continuation” removes training data constructed by “Human-Guided Continuation”. **Bold** indicates the best result.

Model	Coherence	Creativity	Characterization	Fluency	Relevance	Average
STORYREWARD-QWEN	77.5	78.8	71.6	74.2	69.1	75.0
(-) Premise Back-Generation	82.3	70.4	78.6	68.9	75.0	73.9
(-) Human-Guided Continuation	81.5	70.4	76.2	67.1	72.8	72.8
(-) Prompt-Guided Rewriting	73.4	68.1	71.3	65.2	68.7	69.9

The evaluation procedure is as follows. First, we use Llama3.1-8B-Instruct as the policy model to generate 16 candidate stories for each prompt. Then, we use different reward models to select the best story among the 16. Finally, we evaluate these selected stories using the above evaluation method. The results are reported in Table 10. We can find that stories selected by both StoryReward-Qwen and StoryReward-Llama achieve a higher score, which demonstrates the effectiveness of our trained reward models.

H.4 LINGUISTIC ANALYSIS

We conduct linguistic analysis on stories selected by different reward models. Specifically, we adopt the concept of linguistic burstiness from quantitative linguistics to assess human-like writing. Human language is known to exhibit non-uniformity in information content and structure (Church & Gale, 1995; Katz, 1996). Following prior research (Roberts, 1996; Grieve, 2007), we use sentence-length distribution as a proxy for this phenomenon. Specifically, we quantify story burstiness via the kurtosis of sentence lengths. Its calculation formula is:

$$\text{Kurtosis} = \frac{E[(X - \mu)^4]}{\sigma^4} - 3 \quad (6)$$

μ denotes the average sentence length. σ denotes the standard deviation of sentence length. High kurtosis indicates a frequent presence of extremely short or long sentences, which captures the structural complexity and dynamic variance typical of human writing. The results are shown in Table 8. We observe that stories selected by STORYREWARD-QWEN exhibit higher kurtosis compared to the ground truth references. This suggests a potential bias: since the training data includes human-written stories, STORYREWARD-QWEN may prefer linguistic burstiness (i.e., high kurtosis) as a proxy for quality. While this provides a statistical explanation for the model’s behavior, a systematic and in-depth analysis (e.g., mechanistic interpretability) requires substantial experiments beyond the scope of this paper. We leave it as future work.

Algorithm 1: Four-Stage Dimension Selection. This algorithm introduces how we determine which preference dimension the story belongs to.

Function CategorizeDimension($M, \tau, \text{priority}$):

```

// Stage 1: highest mean gaps
best ← score[chosen_story];
ANS ← dim | max(best[dim] − M.mean[dim]) <  $\tau$ ;
if ANS then
  | return ANS
end
// Stage 2: highest margins among Stage 1 candidates
second = M.max;
ANS ← dim | max(best[dim] − second[dim]) <  $\tau$ ;
if ANS then
  | return ANS
end
// Stage 3: lowest rejection variance among Stage 2 candidates
ANS ← dim | |M.rej_vars[dim]| <  $\tau$ ;
if ANS then
  | return ANS
end
// Stage 4: use predefined priority to break ties
foreach dim in priority do
  | if dim.max == M.max then
    | | return dim
  | end
end

```

Table 3: Example prompts for constructing preference pairs by contrasting stories generated by small-scale versus large-scale LLMs.

Prompt Type	Example
How we generate premise based on two stories	Write a continuous, uninterrupted, highly detailed literary fiction story of at least {length} words, with no chapters, no scene breaks, and no meta commentary. Maintain a consistent atmospheric tone, vivid sensory descriptions, and deep psychological introspection. Premise:{prompt} The story should: - Flow continuously without numbered sections or chapter titles - Use rich imagery and internal monologue to convey the protagonist’s unraveling thoughts - Gradually escalate tension without giving definitive answers to the mystery - End with an ambiguous but emotionally powerful conclusion
